PLATFORM / ARCHITECTURE / PROVENANCE / CASE STUDY

CASE STUDY

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COMPANY HUB

A small UK firm wants every contract, meeting and email traceable to the others in a briefing.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A small UK professional services firm holds its commercial knowledge in signed contracts, meeting notes, email threads and shared documents. Before each client call someone rebuilds the picture by hand. Obligations are missed and nobody can show where a statement came from.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Managing director**	Needs a trusted briefing before each client call without hours of preparation.
02	**Contracts lead**	Must know which obligations and dates apply and where each came from.
03	**Data protection lead**	Must show lawful use of personal data and control who can see what.

SCOPE AND HONEST LIMITS

- ⚠ Everything here is a proposed design tested on synthetic data and no results are claimed.
- ⚠ Tamper evidence shows a change occurred and does not prove the original was accurate.
- ⚠ Entity resolution will make mistakes so human review remains part of the workflow.
- ⚠ A real deployment needs legal review and a data protection impact assessment.

CURRENT WORKFLOW AND FAILURE POINTS

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PROPOSED RESEARCH / NOT YET VERIFIED

HOW THE WORK RUNS TODAY

1 TODAY

Request a briefing

A director asks a colleague to gather everything on a client before a call.

2 TODAY

Search several places

The colleague searches mail and shared folders and the meeting notes separately.

3 TODAY

Open each document

Contracts and notes are read by eye to find dates and promises and open actions.

4 TODAY

Copy into a summary

Key points are pasted into a note with no link back to the passage they came from.

5 TODAY

Resolve names by memory

The same client appears under several names and the colleague matches them from memory.

6 TODAY

Share the note

The note is emailed as a copy that goes stale once any source changes.

WHERE IT BREAKS

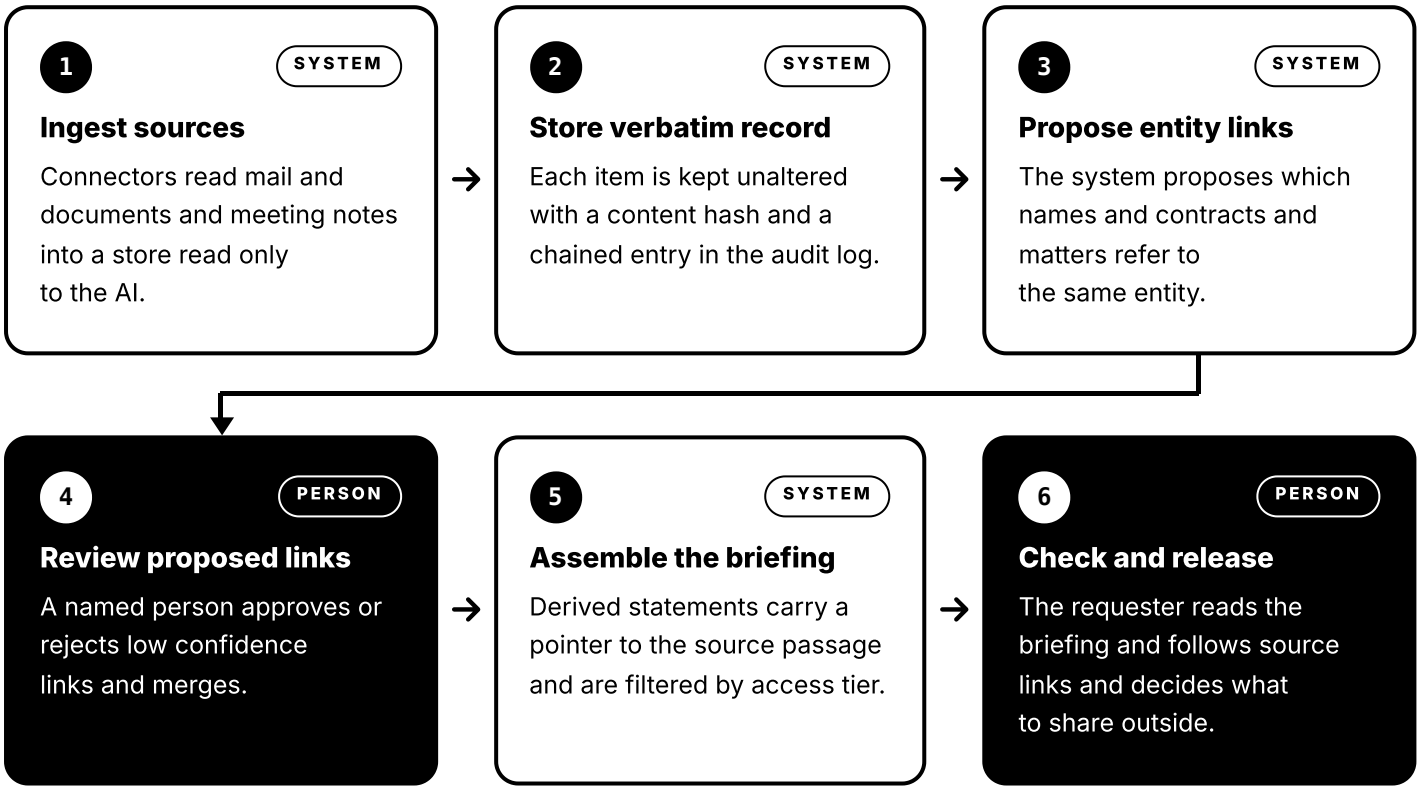
No traceability	A claim in the note cannot be checked against the exact sentence it came from.
Missed obligations	A date or promise buried in an old email thread never reaches the briefing.
Duplicate identities	One client or person is split across several spellings and records.
Uncontrolled sharing	Copied notes can carry restricted content to people who should not see it.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

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PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW



SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Merge two entities into one	RECORDS OWNER	A wrong merge mixes two clients and spreads that error into later briefings.
Share a briefing outside the firm	DIRECTOR	Only a person can judge commercial and legal sensitivity of what leaves.
Widen an access tier	DATA PROTECTION LEAD	Access to personal data needs a lawful basis that a person must own.

ARCHITECTURE, DATA AND CONTROLS

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PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM ARCHITECTURE

1	**Source integration**	Read only connectors to Google Workspace mail and documents and calendar items.
2	**Provenance store**	Append only verbatim records with hashes kept separate from derived interpretation.
3	**Entity resolution**	Matching of people and organisations and contracts with scores and a review queue.
4	**Briefing engine**	Anthropic API reasoning that writes sourced statements and refuses unsupported ones.
5	**Access and audit**	Role based access with organisational tiers and a tamper evident log of every read.

CORE DATA ENTITIES

- **Source item**
- **Verbatim record**
- **Derived statement**
- **Person**
- **Organisation**
- **Contract**

RISKS AND THE CONTROL FOR EACH

Summary replaces the original →	Verbatim records are immutable and derived text always links to its source passage.
Silent alteration of a record →	Content hashes and a hash chained log let any change be detected and reported.
Restricted content in a briefing →	Access tiers are applied before retrieval so the model never sees unpermitted text.
Wrong entity merge →	Merges below a confidence threshold wait for human approval and can be reversed.

EVALUATION AND DELIVERY PLAN

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PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Claim traceability	Share of briefing statements that resolve to a real source passage on review.
Entity match quality	Precision and recall of proposed links against labelled synthetic ground truth.
Obligation recall	Share of planted obligations and dates that appear in the relevant briefing.
Access leakage	Count of restricted passages that surface in a briefing for an unpermitted role.
Preparation effort	Time a reviewer needs to prepare and verify a briefing compared with the manual route.

TEST PLAN

1. **Replay a synthetic firm with planted duplicates and check that links match the ground truth.**
2. **Alter one stored record and confirm the audit check reports the exact break.**
3. **Query as each role with adversarial prompts and confirm restricted text never appears.**
4. **Ask for a statement with no source and confirm the engine declines to write it.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree the briefing questions and the record types and the people who own each.	Map where contracts and meetings and emails live and how names vary across them.	Define the provenance model and access tiers and the data protection assessment.	Build connectors and the store and matching and the briefing engine on synthetic data.	Pilot with a small group and review traceability and leakage results before wider use.
DECISION Which briefings matter most	**DECISION** Which sources are in scope	**DECISION** Access tier definitions	**DECISION** Link approval thresholds	**DECISION** Whether to widen the pilot

DELIVERABLES

- Provenance data model specification
- Access tier and role policy
- Synthetic firm dataset with ground truth links
- Briefing traceability test report
- Data protection assessment template